

	frequently, thus increasing the resistance.	
7(a)	The magnetic flux through any surface area is defined as the <u>product of the component of the magnetic flux density normal to the plane of the surface and the area of the surface.</u>	
7(b)	The suspended magnet is first repelled upwards as the dropped magnet approaches the left coil. As the left magnet approaches the left coil, there is a change of flux linkage associated with the coil. By Faraday's law of EMI, there will be an induced e.m.f.. This induced e.m.f. will cause a current to flow in the closed loops (of the left and right coils). This induced current will, by Lenz's law, flow in such a direction as to oppose the change causing it, i.e. it will be such that the top end of the left coil will be a South pole. This direction of current flow will cause the top end of the right coil to become a South	

	pole too. This causes a momentary repulsion of the suspended magnet. As the dropped magnet leaves the left coil, the suspended magnet is attracted downwards. As the dropped magnet leaves the left coil, the bottom end of the left coil becomes a South pole according to Lenz law. This means the top ends of both coils become a North pole as explained above. An attraction of the suspended magnet downwards is observed.	
7(c)	With switch S opened, the <u>suspended magnet will not move</u> as although there are still induced e.m.f. on the left coil, there is <u>no induced current in the right coil and the right coil is thus not magnetised</u>.	
8(a)	Electric field strength at a point is the electric force per unit	